# A FIELD EXTENSION WHOSE MULTIPLICATION MAP IS FLAT IS ALGEBRAIC SEPARABLE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- If L tensor_K L -> L is flat then for every intermediate field K subset L' subset L the map L' tensor_K L' -> L' is flat. invoked at proof lines 30494-30495. Allows passing to finitely generated subextensions.
- A finitely generated field extension which is formally unramified is finite separable. invoked at proof lines 30497-30500. Concludes the reduced case.

*Source reference.* Original lemma 092K: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30374–30374. Cited at proof lines 30494-30495 to descend the flatness hypothesis to intermediate fields; the conclusion is used inside the counted span.

*Source reference.* Original lemma 092M: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30417–30417. Cited at proof lines 30496-30497 for the formal unramifiedness of L/K; the conclusion is used inside the counted span.

*Source reference.* Original lemma 090W: sources/stacks-chow/tags.txt, source lines 11424–11424. Cited at proof lines 30499-30500 to upgrade formal unramifiedness to finite separability; the conclusion is used inside the counted span.

**Lemma 0.2.** *Let $L/K$ be an extension of fields. If $L \otimes_K L \to L$ is flat, then $L$ is an algebraic separable extension of $K$.*

*Proof.* By Lemma 092K we see that any subfield $K \subset L' \subset L$ the map $L' \otimes_K L' \to L'$ is flat. Thus we may assume $L$ is a finitely generated field extension of $K$. In this case the fact that $L/K$ is formally unramified (Lemma 092M) implies that $L/K$ is finite separable, see Algebra, Lemma 090W. □

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